

House Price Prediction using Machine Learning

Project Summary

This project aims to predict residential house prices using Machine Learning techniques. The dataset contains 545 records and 13 features, including area, bedrooms, bathrooms, stories, parking availability, furnishing status, and other amenities.

Data Preprocessing

- Loaded and explored the Housing dataset.
- Checked for missing values and data quality issues.
- Converted categorical variables into numerical format using encoding techniques.
- Prepared the dataset for machine learning model training.

Exploratory Data Analysis (EDA)

Several visualizations were created to understand the data:

- House Price Distribution Histogram
- Correlation Heatmap
- Area vs Price Scatter Plot

Key observations:

- House price has a strong positive relationship with property area.
- Number of bathrooms significantly affects house price.
- Air conditioning, parking availability, and preferred area also contribute to higher prices.
- Features such as hot water heating and basement have comparatively lower impact.

Machine Learning Models

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regression

The models were assessed using standard regression metrics including:

- R^2 Score
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

Feature Importance Findings

The most influential features affecting house prices were:

1. Area
2. Bathrooms
3. Air Conditioning
4. Stories
5. Parking Availability
6. Preferred Area

Business Insights

For sellers, increasing usable area, improving property facilities, and ensuring parking availability can increase property value. Buyers should pay close attention to property area, number of bathrooms, location quality, and available amenities when evaluating a house.

Conclusion

The analysis demonstrates that house characteristics such as area, bathrooms, parking, and air conditioning strongly influence residential property prices. Machine Learning models can effectively estimate house prices and provide valuable insights for buyers, sellers, and real estate companies.